# Identification of potent inhibitors for blocking the interaction between HIV-1 IN-LEDGF/p75 using virtual screening and docking studies

**Umesh Panwar** and Sanjeev Kumar Singh*

*Computer-Aided Drug Design and Molecular Modeling Lab, Department of Bioinformatics, Alagappa University, Karaikudi-630004, Tamil Nadu, India*

*E-mail: umesh.panwar07@gmail.com, skysanjeev@gmail.com**

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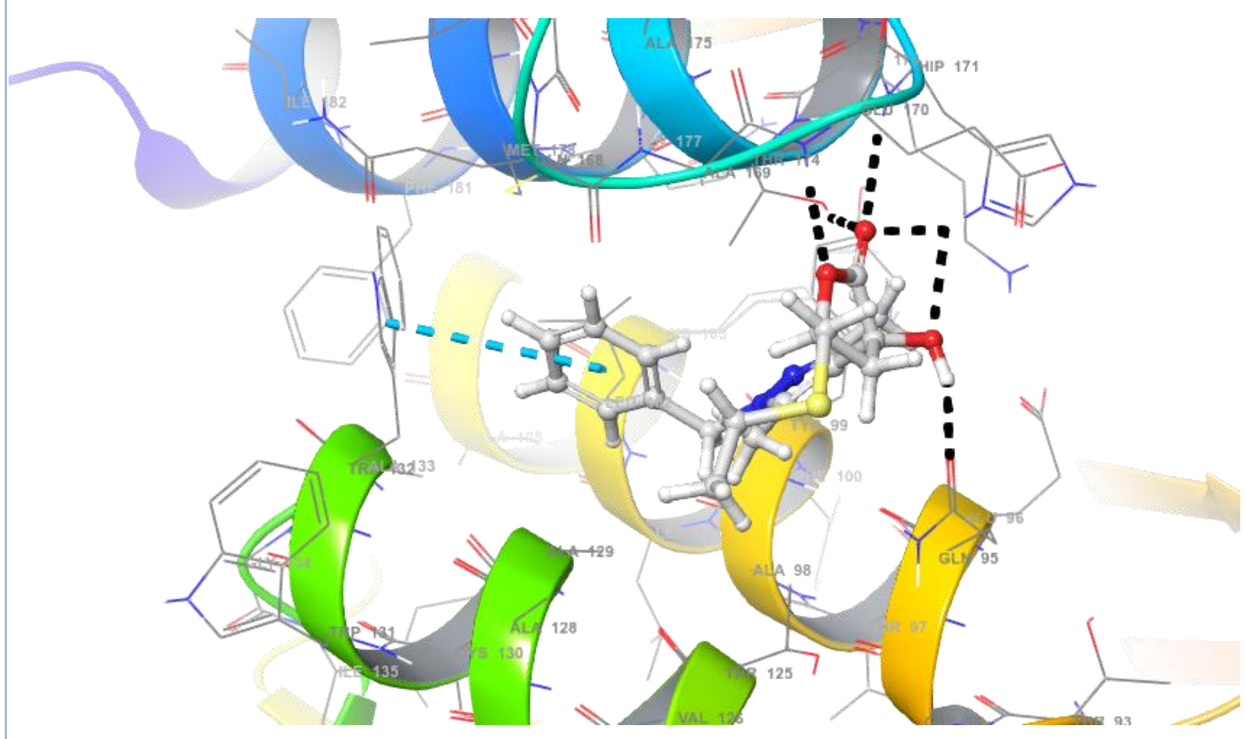
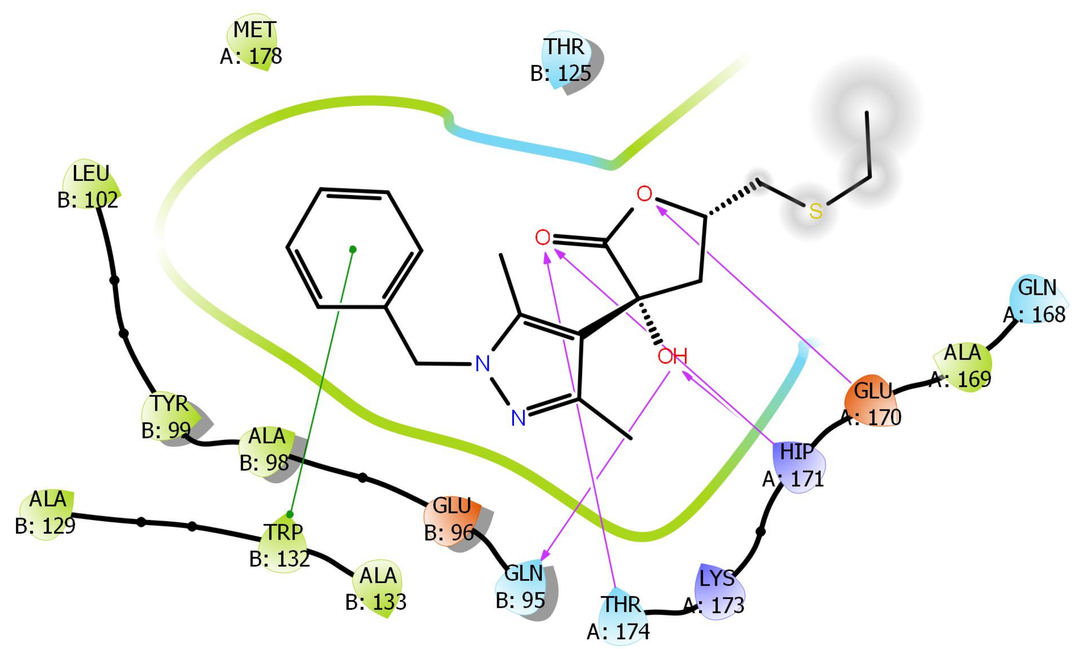
**Abstract**

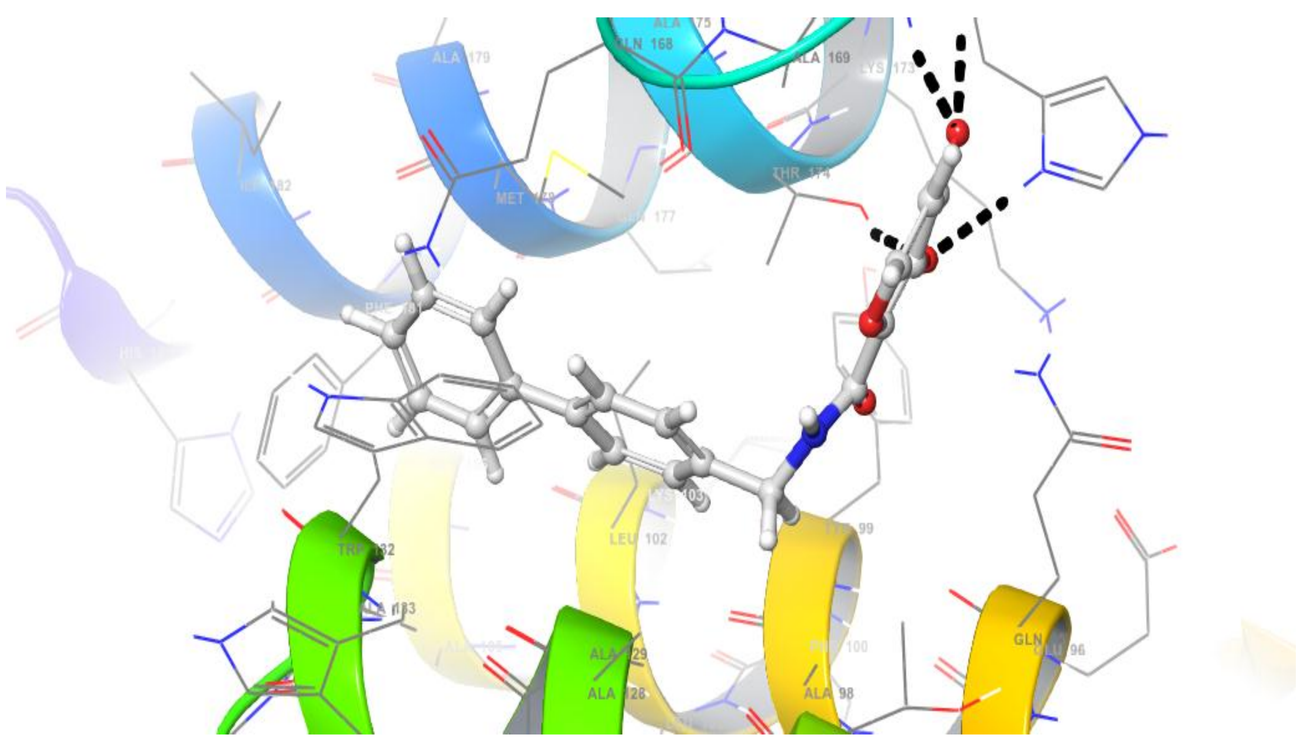
Integrase is an essential enzyme that processes the integration of reverse transcribed viral cDNA into the host genome. To accomplish a successful infection, the role of cellular cofactor proteins, specifically Lens Epithelium-Derived Growth Factor (LEDGF/p75) is necessary during replication cycle. The protein-protein interaction of IN-LEDGF/p75 has become a promising drug target in the drug discovery. Therefore the present study designed to identify a novel compound using *in silico* approaches. Resulted outcome suggest that the compounds NCI 744471 and BindingDB_27849 are found with enhanced binding affinity to inhibit the interaction between IN and LEDGF/p75. Overall, the obtained leads could be therapeutic agent to prevent and treat the HIV-1 viral infection.

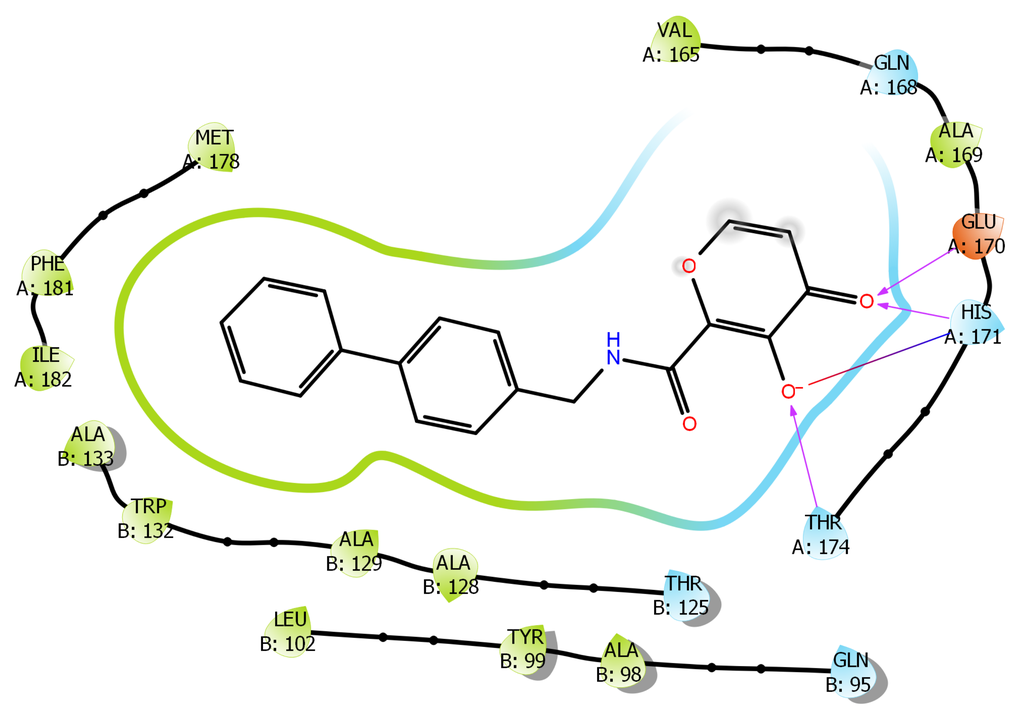
**Keywords:** HIV-1 integrase, LEDGF/p75, Virtual screening, Docking.






## NCI 744471






## BindingDB_27849

**The 2D and 3D view of Screened Hits against HIV-1 IN-LEDGF/p75 interaction**

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